

ASHWINI KUMAR SHARMA

SENIOR DATA ENGINEER | DATA PLATFORM | DATA RELIABILITY

aksharma2004@gmail.com | +91-9547973377 | linkedin.com/in/aksharma2004 | Siliguri, West Bengal, India

PROFESSIONAL SUMMARY

Senior Data Engineer with 15+ years of experience designing, building and operating production-grade data platforms, ingestion systems, ELT/ETL pipelines and analytical data products across AWS and GCP. Hands-on with Python, SQL, Snowflake, BigQuery, Redshift, Databricks, dbt, Airflow, Dagster and Prefect, with strong experience in reliability, data quality, observability, lineage, governance, performance and cloud cost optimisation. Proven record of diagnosing production failures, creating repeatable recovery practices, reducing technical debt and replacing manual or vendor-based workflows with scalable engineering solutions. Experience supporting AI/LLM use cases, including RAG data infrastructure and AI-assisted engineering workflows, while partnering with business and technical stakeholders in remote, cross-functional teams.

CORE SKILLS

Data Engineering: Python, SQL, PySpark, Scala, Java, ETL/ELT, batch pipelines, streaming/batched ingestion, API/GraphQL integrations

Warehouses & Platforms: Snowflake, BigQuery, Amazon Redshift, Databricks, data lakes, OLAP/OLTP

Transformation & Orchestration: dbt Core/Cloud, Apache Airflow, Dagster, Prefect, Fivetran, Airbyte, DLTHUB

Reliability & Data Quality: Data quality assurance, monitoring, alerting, anomaly detection, freshness, pipeline health, lineage, incident RCA, runbooks, SLAs

Cloud & Infrastructure: AWS S3/EC2/Lambda, GCP GCS/GKE, Terraform, Ansible, Docker, Kubernetes, Git, CI/CD

Modelling & Governance: Kimball dimensional modelling, schema design, RBAC, permissions, data retention, column-level lineage, governance

Engineering Practices: Git-based development, CI/CD automation, code review, automated testing, deployment and production release practices

AI / Data for AI: RAG data infrastructure, chatbot analytics, LLM-assisted engineering, Cursor, Claude, MLOps/DataOps practices

Analytics: Sigma Computing, Looker, Power BI, KPI-oriented self-service datasets

Leadership: Technical mentoring, stakeholder management, Agile/Scrum, architecture, technical debt management, remote collaboration

PROFESSIONAL EXPERIENCE

Senior Data Engineer (Contract) | Uplearn UK | Jul 2026 – Present

- Build and maintain dbt models supporting Uplearn's data platform; improve production model efficiency through partitioning and incremental processing.
- Created a BigQuery data-retention policy to reduce unnecessary warehouse data, improve query efficiency and control platform cost.
- Generate models to Reverse ETL into Hubspot
- Work in a remote, production-focused environment with emphasis on maintainability, performance and reliable data delivery.

Applied Git-based development and CI/CD practices to dbt and data-platform changes, supporting review, testing and controlled production releases.

Senior Data Engineer (Contract) | Brainforge, USA | Dec 2025 – Jul 2026

- Designed and operated a GraphQL-based ELT pipeline ingesting 20M+ SPINS records (~30 GB) monthly into Amazon Redshift, orchestrated with Prefect Cloud; eliminated 40 hours/month of manual Excel processing.
- Built a greenfield Snowflake warehouse for the Consumer Technology Association, including RBAC, warehouse objects, S3/Salesforce Marketing Cloud and Shopify ingestion, and dbt dimensional modelling.
- Implemented rule-based and fuzzy-match identity resolution for 100K+ event registrations, turning inconsistent event/exhibitor data into an analytics-ready entity layer.
- Configured Metaplane observability for BigQuery datasets, monitoring freshness, dbt pipeline health, row-count and column-level anomalies to detect failures proactively.
- Built an end-to-end influencer data pipeline from Upfluence via Segment into BigQuery, dbt transformation and reverse ETL into Northbeam using Dagster; replaced manual Google Sheets tracking across ~50 influencers and thousands of monthly orders.
- Implemented Kimball-style dimensional models across Snowflake, BigQuery and Redshift and delivered KPI-aligned self-service datasets and dashboards in Sigma Computing.

Applied Git-based development and CI/CD practices across client data-engineering projects, including code review, automated testing/deployment and controlled production releases.

Senior Data Engineer (Contract) | Haven Holidays – Bourne Leisure, UK | May 2025 – Dec 2025

- Diagnosed and resolved production data-integrity issues across Coupa procurement ingestion and dbt models, reconciling millions of Purchase Orders and ~£40M annual procurement spend against a legacy ERP.
- Reverse-engineered undocumented VAT logic from legacy SQL Server and rebuilt the logic in Snowflake/dbt, achieving exact reconciliation with legacy tax reports and establishing Snowflake as the authoritative reporting source.
- Strengthened modelling, pipeline reliability, documentation and remote async engineering practices.

Senior Data Architect | BSI Financial (formerly Entra Solutions Pvt Ltd) | Jun 2023 – Dec 2025

- Led end-to-end modernisation from a legacy SQL Server warehouse to Databricks + Snowflake, consolidating 8 heterogeneous sources including COBOL/SFTP, SQL Server, MongoDB, Postgres, Jira, Google Sheets and S3 into a dbt-modelled dimensional platform.
- Replaced three vendor solutions with in-house data engineering, delivering \$135K in annual savings and reducing dependence on brittle/manual transformation workflows.
- Built analytics on customer-chat data showing 75% chatbot query resolution, providing the evidence for a customer-service operating-model change.
- Modelled Five9 call-centre event/log data to identify demand patterns and improve agent scheduling and service performance.
- Built a collection-agent performance scoring model using operational data to support objective performance management and collections improvement.
- Designed the schema and data infrastructure for a RAG-based customer-service chatbot, enabling AI-powered workflows at scale.
- Implemented data observability and column-level lineage using Datadog; administered Databricks and Snowflake roles and permissions through Terraform.
- Developed a DSL that allowed business-authored Excel transformations to be compiled into PySpark, reducing the translation gap between business and engineering teams.
- Drove architecture, technical debt reduction and cross-functional delivery across Finance, Collections, Call Centre and Customer Service domains.
- Built data pipelines for mortgage servicing time-series data using daily source-system snapshots, ingesting and transforming successive snapshots to maintain historical views of loan balances, servicing activity and portfolio changes.
- Designed warehouse models to support day-over-day analysis and historical portfolio tracking, with data quality and reconciliation checks across ingestion and transformation layers.
- Worked with large volumes of longitudinal mortgage data where timeliness, completeness and consistency of each daily snapshot were critical to downstream analytics.
- Applied Git/CI/CD practices across data-platform engineering, enabling version-controlled transformations, repeatable deployments and safer production releases.

Data Reliability Engineering Manager | Shopify | Apr 2022 – Jun 2023

- Performed daily root-cause analysis across production data pipelines and models running on GKE + Apache Airflow; authored runbooks for recurring cluster-level failures to make incident recovery repeatable.
- Built end-to-end data lineage and dataset ownership mapping so failed pipelines could be traced to affected downstream models and the correct owners notified quickly.
- Analysed GCS storage usage and lifecycle patterns, archiving/deleting stale data to deliver \$80K annual savings.
- Managed technical debt in a Trino-based analytical platform and implemented Terraform-based Infrastructure as Code practices across data services.

Used Git and CI/CD workflows for data-service and infrastructure changes, improving release repeatability and reducing deployment risk.

Engineering Manager – Connectors | Fivetran | Oct 2018 – Mar 2022

- Led 3 teams of 15 engineers responsible for 20+ production ELT connectors across eCommerce, CRM and HR systems including Salesforce, Shopify, Stripe, NetSuite, SAP Concur, Microsoft Dynamics 365 and Xero; maintained 99.9% connector uptime.
- Redesigned high-cost BigQuery log ingestion from real-time inserts to batched ingestion, reducing cost from ~\$1,000/day to ~\$200/day and saving approximately \$292K annually.
- Delivered customer-facing pipeline logs exposing extraction start/end times, API request details, row counts and data volumes, enabling downstream automation from log events.
- Resolved a chronic NetSuite connector integrity issue with checksum-based source/target verification, reducing support incidents and restoring customer confidence.
- Improved engineering practices and reduced system errors by 40%; introduced backward sync to reduce data availability lag by weeks.
- Led teams operating Git-based engineering and CI/CD workflows for production ELT connectors, supporting disciplined code review, automated testing and reliable releases.
- Designed and implemented handling for late-arriving records across Fivetran connectors, ensuring delayed source updates were correctly captured and reflected in downstream warehouse tables without data loss or duplication.

- Built and supported reprocessing and backfill mechanisms for warehouse data, enabling historical records to be safely re-ingested and reconciled when source data changed or ingestion issues occurred
- Investigated data discrepancies and connector failures, using source-to-warehouse validation and checksum/reconciliation techniques to maintain data integrity.

Principal Member of Technical Staff | Oracle | Jun 2013 – Oct 2018

- Designed and optimised REST adapters for Oracle SOA Suite, including authentication and throughput improvements; led JMS adapter performance work and Enterprise Delivery Network operations.
- Migrated build pipelines from ANT to Maven, improving build reproducibility and engineering efficiency.

Tech Lead | Thomson Reuters Pvt Ltd | Jun 2009 – May 2013

- Built and maintained the Novus Boolean Search indexing solution processing petabytes of legal data; developed wildcard-search indexing and performance improvements.
- Mentored a team of 8 engineers and supported delivery and technical standards.

Tech Lead | Wipro Technologies | Jul 2005 – Jan 2009

- Designed and implemented web-based solutions for Novartis Pharmaceuticals, including Course Tracking, People Search and expenditure-tracking solutions.

SELECTED IMPACT

- \$135K annual savings at BSI Financial by replacing three vendor solutions with in-house engineering.
- ~\$292K annual savings at Fivetran through an 80x reduction in BigQuery log-ingestion cost.
- \$80K annual GCS storage savings at Shopify through data lifecycle optimisation.
- 40 hours/month of manual work eliminated at Magicspoon through automated 20M+ monthly SPINS ingestion.
- 100K+ event registrations resolved through identity stitching at CTA.
- Exact reconciliation of £40M procurement spend and complex VAT logic at Haven Holidays.
- 99.9% uptime maintained across 20+ production ELT connectors at Fivetran.

EDUCATION

- Master of Technology (M.Tech), Computer Science — BITS Pilani, 2015
- Bachelor of Technology (B.Tech), Computer Science — Sikkim Manipal University, 2003